Windows :

FOCA

Nessus

Sysmon

Maltego

Kali :

whois

nslookup

Nmap

armitage

Theharvester

Netdiscover

Sublist3r

Wireshark

Browser:

Robtex

Shodan

Google Dorking

Armitage usage in kali linux:

1. sudo msfdb init

2. armitage

sysmon will not show most of the events unless you configure it.

1. Create an xml file with the given content

2. cmd as administrator

3. sysmon64.exe -c <file-name>.xml

```
<Sysmon schemaversion="4.90">
  <EventFiltering>
    <NetworkConnect onmatch="exclude" />
    <FileCreate onmatch="exclude" />
    <RegistryEvent onmatch="exclude" />
    <DnsQuery onmatch="exclude" />
    <FileDelete onmatch="exclude" />
    <FileDeleteDetected onmatch="exclude" />
  </EventFiltering>
```

</Sysmon>

Ethical Hacking Lab Exam — Master Practical Guide

Experiment 1 — Setting up the virtual cybersecurity lab

Kali Linux Installation_Guide.pdf

Topic: Kali Linux, Metasploitable 2, Windows XP, VirtualBox, and network connectivity.

Aim

To set up a virtual cybersecurity laboratory and verify communication between virtual machines.

What each tool is for

Oracle VM VirtualBox

Runs virtual computers inside your main computer.

Kali Linux

The system you use to run cybersecurity tools.

Metasploitable 2

An intentionally vulnerable Linux machine for authorized practice.

Windows XP

The Windows virtual machine used in the manual to practise basic network commands.

Step 1: Open VirtualBox

1. Start your lab computer.
2. Open the Windows Start menu.
3. Search for Oracle VM VirtualBox.
4. Open it.
5. Find the virtual machines listed in VirtualBox.

If the lab machines are already configured, you do not need to install them again.

Step 2: Start Kali Linux

1. Select the Kali Linux virtual machine.
2. Click Start.
3. Wait for the operating system to load.
4. Log in if prompted.
5. Open Terminal from the desktop or Applications menu.

You should now have a terminal ready for commands.

Step 3: Find Kali Linux's IP address

Type:

```
ip a
```

Press Enter.

What this does:

- `ip` is the Linux networking command.
- `a` means addresses.
- The output lists network interfaces and their assigned IP addresses.

Look for an interface such as `eth0` or `enp0s3` and an address shown after `inet`. Ignore `127.0.0.1`, which is the loopback address.

Step 4: Start Metasploitable 2

1. Return to the VirtualBox Manager.
2. Select Metasploitable 2.
3. Click Start.
4. Wait for the login prompt.
5. Use the lab credentials provided in your manual: `msfadmin` for both username and password.
6. Log in.

Step 5: Find Metasploitable 2's IP address

At its Linux terminal, type:

```
ip a
```

Press Enter.

Find the active network interface and note its IPv4 address. This is the address you'll use when testing connectivity from Kali.

Step 6: Test connectivity from Kali

Return to the Kali terminal. Type the following, replacing the placeholder with Metasploitable 2's actual IP address:

```
ping <Metasploitable-IP>
```

For example, if the assigned address is `192.168.56.10`, the command would be:

```
ping 192.168.56.10
```

Press Enter.

Expected output: If the machines can communicate and the target responds to ping, you should see replies containing information such as `bytes from`, `icmp_seq`, and `time`.

Stop the continuous ping using Ctrl+C.

If there are no replies, check that both virtual machines are running and their network adapters are configured on a compatible lab network. A missing ping response alone does not prove that the machine is offline.

Step 7: Find the Windows XP IP address

1. Start the Windows XP virtual machine in VirtualBox.
2. Log in if prompted.
3. Click Start → Run.
4. Type `cmd`.
5. Press Enter.
6. In the Command Prompt, type:

```
ipconfig
```

7. Press Enter.
8. Note the IP address shown for the active network connection.

Step 8: Understand the networking modes

Your setup manual discusses virtual-machine networking, including NAT and Host-Only networking.

- NAT: A virtual machine can access external networks through the host's network connection, depending on configuration.
- Host-Only: Virtual machines can communicate through a virtual network isolated from the external network by default.

- IP address: A numerical address assigned to a network interface.
- Ping: A utility used to test IP connectivity using ICMP echo requests and replies.

For the lab, use the network mode specified by your instructor so Kali can reach the assigned target.

Result

Kali Linux, Metasploitable 2, and Windows XP were started in VirtualBox. Their IP addresses were identified, and connectivity was tested using the appropriate commands.

Experiment 2 — Reconnaissance and system monitoring

Week2_practical.pdf

Topics: Sysmon, theHarvester, Netdiscover, and WHOIS.

A. Sysmon — System monitoring on Windows

Sysmon is a Windows system-monitoring utility that records selected system activities in event logs.

Step 1: Open Windows

1. Start the Windows machine specified in your lab.
2. Open the folder containing `sysmon.exe`.
3. If your lab provides a Sysmon configuration file, locate `sysmonconfig.xml` in the appropriate folder.
4. Open Command Prompt as Administrator.

Step 2: Check the Sysmon commands

Navigate to the folder containing `sysmon.exe`, or use the full path to the executable.

The commands in your PDF include:

```
sysmon.exe -?
```

Displays help information.

```
sysmon.exe --version
```

Displays the version if supported by that Sysmon build.

```
sysmon.exe -c
```

Displays the current configuration.

Step 3: Install Sysmon if it is not already installed

From the directory containing the executable, run:

```
sysmon.exe -i
```

This installs Sysmon with its default configuration.

If your instructor provides a configuration file, use:

```
sysmon.exe -i sysmonconfig.xml
```

To apply or update a configuration, your manual also lists:

```
sysmon.exe -c sysmonconfig.xml
```

To uninstall Sysmon, the manual lists:

```
sysmon.exe -u force
```

Important: Run installation and removal commands only when required. If Sysmon is already configured in the lab, do not reinstall it unnecessarily.

Step 4: View the logs

1. Press Windows + R.
2. Type `eventvwr.msc`.
3. Press Enter.
4. Open Applications and Services Logs → Microsoft → Windows → Sysmon → Operational, if that log is available.
5. Examine the recorded events.

Result: Sysmon is configured to record the system activities selected by its configuration.

B. theHarvester — Public information gathering

theHarvester is an OSINT tool used to find publicly available information about a domain, such as email addresses, subdomains, hostnames, and IP addresses.

Step 1: Open Kali Terminal

Open the Kali Linux virtual machine and click the Terminal icon.

Step 2: Check whether theHarvester is installed

```
theHarvester --help
```

If help information appears, the command is available.

Step 3: Install it if necessary

```
sudo apt update
```

Wait for the package list to update, then run:

```
sudo apt install theharvester
```

Enter your password if prompted.

Step 4: Run a basic search

Your PDF shows this example:

```
theHarvester -d microsoft.com -b all -l 300 -q
```

Meaning:

- `-d` specifies the domain.
- `-b` specifies the data source or sources.
- `all` requests the supported sources available in that version.
- `-l 300` limits the number of results.
- `-q` requests quiet output.

Use the domain assigned by your instructor when the exercise requires a specific target. Results depend on available sources and current service support.

Result: Publicly available domain-related information is collected and displayed.

C. Netdiscover — Local network host discovery

Netdiscover helps identify active devices on a local network using ARP discovery.

Step 1: Open Kali Terminal

Step 2: Check help

```
netdiscover -h
```

Step 3: Start discovery

```
sudo netdiscover
```

Allow the tool to discover hosts on the applicable local network.

Step 4: Scan a specified lab range

Your PDF provides examples such as:

```
sudo netdiscover -r 192.168.1.0/24
```

Replace the example range with the network range authorized for your lab.

You can also specify an interface:

```
sudo netdiscover -i eth0
```

Other options listed in your PDF include:

```
sudo netdiscover -p
sudo netdiscover -S
```

These change Netdiscover's operating mode or output behavior; consult `netdiscover -h` for the options supported by the installed version.

Expected output: A list of discovered hosts, potentially including IP addresses, MAC addresses, and vendor information.

D. WHOIS — Domain registration lookup

WHOIS retrieves publicly available registration information for domains and IP addresses.

Step 1: Open Kali Terminal

Step 2: Verify the installation

```
whois --version
```

Step 3: Query a domain

```
whois example.com
```

Look for available registration details, registrar information, registration dates, status, and nameservers.

Step 4: Query an IP address

```
whois 8.8.8.8
```

This looks up registration or network-allocation information associated with the IP address.

Step 5: Query a specific WHOIS server

```
whois -h whois.verisign-grs.com example.com
```

Here, `-h` specifies the WHOIS server to query.

Result: Registration and network-allocation details are retrieved where publicly available.

Experiment 2 — Result

Sysmon was used to examine system monitoring, while theHarvester, Netdiscover, and WHOIS were used to gather publicly available domain information and discover local hosts.

Experiment 3 — WHOIS, nslookup, theHarvester, DNS mapping, and Nikto

Week-4 Nikto.pdf

Your uploaded Week 4 Nikto PDF contains several separate practical topics, not only Nikto. Study them in this order.

A. DNS basics

DNS stands for Domain Name System. It translates human-readable domain names into IP addresses and stores other records used for email and name-server discovery.

Record	Purpose
A	Maps a domain to an IPv4 address.
MX	Identifies mail servers.
NS	Identifies authoritative name servers.
TXT	Stores text information, often used for verification and email-security policies.

B. WHOIS

Open Kali Terminal and use the following commands.

1. Update package information, if required

```
sudo apt update
```

2. Install WHOIS if missing

```
sudo apt install whois
```

3. Verify

```
whois --version
```

4. Query a domain

```
whois example.com
```

5. Query an IP address

```
whois 8.8.8.8
```

6. Specify a WHOIS server

```
whois -h whois.verisign-grs.com example.com
```

These commands are listed in the WHOIS section of your PDF.

C. nslookup — DNS lookup

`nslookup` queries DNS servers for domain names, IP addresses, and selected DNS records.

Step 1: Install if necessary

```
sudo apt update
sudo apt install dnsutils
```

Step 2: Verify that it works

```
nslookup
```

This opens an interactive prompt. Type `exit` to leave it.

Step 3: Look up a domain

```
nslookup example.com
```

Examine the returned IP address or addresses.

Step 4: Find mail servers

```
nslookup -type=mx example.com
```

Step 5: Find nameservers

```
nslookup -type=ns example.com
```

Step 6: Query a specific DNS server

```
nslookup example.com 8.8.8.8
```

This asks Google's public DNS resolver for the answer.

Step 7: Reverse DNS lookup

```
nslookup 8.8.8.8
```

This attempts to find a name associated with the IP address.

Result: DNS records are queried and interpreted.

D. theHarvester

If you already completed the installation in Experiment 2, you can skip it.

Step 1: Check the version

```
theHarvester --version
```

Step 2: Search a domain using a specified source

```
theHarvester -d example.com -b google
```

Your PDF uses this as an example of searching for publicly listed email addresses.

Step 3: Search certificate-transparency information

```
theHarvester -d example.com -b crtsh
```

Step 4: Limit results

```
theHarvester -d example.com -b crtsh,hackertarget -l 20
```

The `-l` option limits the number of results. Supported sources can change between versions, so a source may not work in every installation.

E. DNS mapping

DNS mapping combines results from WHOIS, nslookup, and theHarvester into one domain profile.

1. Run `whois example.com` and note the registrar and nameservers.
2. Run `nslookup -type=mx example.com` to inspect mail servers.
3. Run `nslookup -type=ns example.com` to inspect nameservers.
4. Run theHarvester against the assigned training domain to gather available subdomains and emails.
5. Organize the collected information into a single report.

Result: A consolidated view of the domain's registration, DNS infrastructure, and publicly available related information is prepared.

F. Nikto — Web server vulnerability scanner

Nikto scans a web server for known issues such as outdated software, potentially exposed files, and common misconfigurations.

Step 1: Open Kali Terminal

Step 2: Verify Nikto

```
nikto -Version
```

Step 3: Install if necessary

```
sudo apt install nikto -y
```

Step 4: Identify the target

Use the IP address of the authorized training web server. The manual gives `192.168.56.10` as an example.

Step 5: Run a basic scan

```
nikto -h 192.168.56.10
```

Replace the example address with the actual assigned target. The `-h` option specifies the host.

Step 6: Wait for the scan to finish

Read the findings printed in the terminal. The scan can take several minutes.

Step 7: Save the results

```
nikto -h 192.168.56.10 -o report.txt
```

This writes the output to `report.txt` in the current working directory.

Common options

Option	Purpose
`-h`	Target host
`-p`	Port
`-ssl`	Force SSL/HTTPS
`-o`	Save output to a file
`-Format`	Select an output format
`-Tuning`	Restrict the scan to selected categories

Interpreting the output: A plus sign (+) commonly introduces a finding. Look for server-version disclosures, outdated software warnings, interesting files or directories, and missing security headers.

Result: The authorized web server is scanned, and the findings are examined for potential vulnerabilities and misconfigurations.

Experiment 4 — FOCA and Google Dorking

Week 4 -FOCA Installation.pdf

Topic: Open Source Intelligence (OSINT) and passive reconnaissance.

A. Google Dorking

Google Dorking uses search operators to find publicly indexed information more precisely.

Step 1: Open a browser

1. Start Windows.
2. Open Chrome or Firefox.
3. Go to Google.

No terminal or special installation is required for the Google search portion.

Step 2: Try the search operators

Use the following examples from the techniques covered in your manual:

Operator	Example	Purpose
`site:`	`site:owasp.org`	Restricts results to a domain.
`filetype:`	`filetype:pdf site:owasp.org`	Finds PDFs indexed for a domain.
`intitle:`	`site:owasp.org intitle:"SQL Injection"`	Searches page titles for a phrase.
`inurl:`	`site:docs.aws.amazon.com inurl:iam`	Looks for a term in the URL.
`intext:`	`site:developer.mozilla.org intext:JavaScript`	Searches page text.
`-`	`site:owasp.org -blog`	Excludes results containing a term.
Quotes	`"exact phrase"`	Searches for an exact phrase.
`OR`	`site:owasp.org OR site:owasp.org`	Combines alternatives; use distinct terms or sites in practice.

Use only public information relevant to the assigned exercise. Do not attempt to access information that you are not authorized to view.

B. FOCA — Metadata extraction

FOCA stands for Fingerprinting Organizations with Collected Archives. It helps analyze publicly available documents and their metadata.

Step 1: Open FOCA

1. Start Windows.
2. Open the folder containing `FOCA.exe`.
3. Double-click `FOCA.exe`.
4. Wait for the application to open.

Step 2: Create a project

1. Choose New Project or the equivalent project-creation option.
2. Enter a project name.
3. Enter the authorized training domain.
4. Select a folder where downloaded documents and project files can be stored.
5. Save the project.

Step 3: Search for documents

1. Use the search option in FOCA.
2. Select the document types or search options required by your manual.
3. Run Search All if available.
4. Review the list of publicly indexed documents.

Step 4: Download documents

1. Select the documents needed for the exercise.
2. Choose Download All if appropriate.
3. Wait for the download to finish.

Step 5: Extract metadata

1. Select Extract Metadata or Extract All Metadata.
2. Allow FOCA to process the downloaded files.
3. Open Analyze Metadata.
4. Review the information found, such as document properties, software details, authors, domains, hosts, or email addresses where available.

The precise labels can vary with the FOCA version. Your PDF describes the project, document-search, download, and metadata-analysis workflow.

Week 4 -FOCA Installation.pdf

Result

Google Dorking was used to find publicly indexed information, and FOCA was used to analyze the metadata of publicly available documents.

Experiment 5 — Maltego and Shodan

Week5 - Maltego & Shodan.pdf

Topic: OSINT investigation and visual link analysis.

A. Maltego

Maltego represents information as entities and links between them.

Step 1: Open Maltego

1. Start the computer on which Maltego is installed.
2. Open Maltego Community Edition (CE).
3. Sign in or register if the application requests it.
4. Complete any required initial configuration.

Step 2: Create a graph

1. Select New Graph or File → New Graph.

2. Find the entity palette.
3. Add a Domain entity to the graph.
4. Enter the authorized training domain.

Step 3: Run transforms

1. Right-click the Domain entity.
2. Look for the available Run Transform options.
3. Choose an appropriate DNS-related transform.
4. Run it.
5. Review any new entities and connections added to the graph.

Transforms may discover DNS names, IP addresses, or other related information, depending on the selected transform and available data sources.

Step 4: Expand and interpret the graph

1. Select a discovered entity.
2. Run another relevant transform if required.
3. Observe how the entities connect.
4. Identify meaningful relationships.
5. Record the findings for the report.

Result: A graph representing relationships between the authorized domain and discovered entities is created.

B. Shodan

Shodan is a search engine for information about Internet-connected devices and their exposed services.

Step 1: Open Shodan

1. Open Chrome or Firefox.
2. Navigate to Shodan.
3. Sign in if required by the available search functionality.

Step 2: Search

1. Enter the IP address or other authorized search target specified by your instructor.
2. Submit the search.
3. Review any returned device information, such as observed ports, services, or software banners.
4. Record the findings relevant to the exercise.

Your manual includes `8.8.8.8` as an example IP address. Actual results depend on Shodan's indexed information and should not be assumed in advance.

Step 3: Cross-check findings

Compare relevant public information with the results obtained through Maltego or other OSINT sources. Differences can occur because data sources are collected at different times.

Result

Maltego was used to visualize relationships between entities, and Shodan was used to examine publicly indexed information about Internet-connected devices.

Experiment 6 — Nmap network scanning

NMAP .pdf

Topic: Host discovery, port scanning, OS detection, and service enumeration.

Step 1: Open Kali Terminal

1. Start Kali Linux.
2. Open Terminal.
3. Check whether Nmap is installed.

```
nmap --version
```

If it is missing, install it:

```
sudo apt update
sudo apt install nmap
```

Step 2: Read the help information

```
nmap --help
```

This displays Nmap's available options.

Step 3: Identify the authorized lab network

Use the assigned target IP and lab network range. Your manual describes scanning an authorized DVWA target inside an isolated lab.

Step 4: Discover live hosts

```
nmap -sn 192.168.56.0/24
```

- `-sn` performs host discovery without a port scan.
- `/24` represents a network range.

Replace the example range with the actual authorized lab range.

Expected output: A list of hosts Nmap identifies as up, along with their IP addresses and potentially their hostnames or other discovery information.

Step 5: Perform a basic port scan

```
nmap 192.168.56.10
```

This scans the target's common TCP ports.

Look for port numbers and their reported states, such as `open`, `closed`, or `filtered`.

Step 6: Scan all TCP ports

```
nmap -p- 192.168.56.10
```

The `-p-` option selects all TCP ports.

Step 7: Perform OS detection

```
sudo nmap -O 192.168.56.10
```

The `-O` option attempts to identify the target operating system. It may require elevated privileges and may not identify the OS accurately in every case.

Step 8: Identify services and versions

```
nmap -sV 192.168.56.10
```

The `-sV` option probes open ports to identify the services and, where possible, their versions.

Step 9: Run a combined scan

Your manual lists this combined example:

```
sudo nmap -sS -sV -O 192.168.56.10
```

- `-sS` performs a TCP SYN scan.
- `-sV` attempts service/version detection.
- `-O` attempts operating-system detection.

It also lists:

```
sudo nmap -A 192.168.56.10
```

`-A` enables several advanced detection features. Use it only if required for your assigned lab target.

Step 10: Save the scan results

Nmap supports output options such as:

```
nmap -oN nmap-report.txt 192.168.56.10
```

This saves a normal-format report to `nmap-report.txt`.

Step 11: Analyze the results

1. Identify the live host.
2. Record open ports.
3. Note the services and versions.
4. Record any OS-detection result.
5. Explain which exposed services may need further security assessment.

Result: Nmap was used to discover hosts, scan ports, identify services, and attempt OS detection on an authorized lab target.

Experiment 7 — Nessus vulnerability assessment

Week-7-Nessus.pdf

Step 1: Open Kali Linux and Firefox

1. Start Kali Linux.
2. Open Firefox from the desktop or Applications menu.
3. Click the address bar.

Step 2: Open Nessus

Enter:

```
https://localhost:8834
```

Press Enter.

This opens the local Nessus web interface if the service is installed and running.

Step 3: Log in

1. Enter the Nessus credentials configured for the lab.
2. If the setup or activation screen appears, follow your instructor's existing lab setup instructions.
3. Wait for Nessus to finish loading.

Step 4: Create a scan

1. Click My Scans.
2. Click New Scan.
3. Select Basic Network Scan.
4. Enter a name such as `Lab Scan`.
5. Enter the assigned target IP address.
6. Click Save.

Step 5: Launch the scan

1. Find the saved scan.
2. Click the Launch/Play icon.
3. Wait for the scan to finish.
4. Open the completed scan.

Step 6: Interpret the severity

Severity	Meaning
Critical	Extremely serious security risk
High	Significant security risk
Medium	Moderate security risk
Low	Lower-severity issue
Info	Informational finding

Step 7: Review CVEs

1. Open the completed scan.
2. Click Vulnerabilities.
3. Select a Critical or High finding.
4. Look in See Also or References.
5. Copy the associated CVE identifier.
6. Open CVE.org or NVD.
7. Search for the CVE identifier.
8. Review its description, CVSS information, impact, and patch recommendations.

Step 8: Export the report

1. Open the completed scan.
2. Select Export.
3. Choose an available report format required by your instructor.
4. Save the report if prompted.

Result: The authorized target was assessed for vulnerabilities, findings were classified by severity, CVE references were reviewed, and a report was generated or exported as required.

Experiment 8 — Robtex: Network and DNS intelligence

Week8_lab_Manual Robtex.pdf

Important: Robtex is a website, so you use a browser rather than a Kali terminal.

Step 1: Open Robtex

1. Open Firefox or Chrome.
2. Enter https://www.robtex.com.

3. Press Enter.
4. Wait for the website to load.

Your PDF says no installation or account is required for the basic lookup.

Step 2: Search a domain

1. Find the Robtex search bar.
2. Enter the authorized training domain.
3. Press Enter.
4. Review the domain summary.

Step 3: Examine the summary

Record the resolved IP address or addresses and any available hosting information.

Step 4: Interpret DNS records

Record	Meaning
A	IPv4 address associated with the domain
MX	Mail servers for the domain
NS	Authoritative nameservers
TXT	Text records, often used for verification or email-security policies

Step 5: Explore the graph

1. Find the graph or relationship view, if available.
2. Examine the connections between the domain and its IP addresses.
3. Review any related domains sharing hosting infrastructure.
4. Record relevant observations.

Step 6: Perform an IP lookup

1. Select an IP address found in the domain results.
2. Search for that IP address in Robtex.
3. Review its network range and ASN information, if displayed.

ASN means Autonomous System Number and identifies an autonomous system participating in Internet routing.

Step 7: Cross-reference the findings

Compare relevant results with WHOIS, crt.sh, or DNSdumpster, as suggested by your manual.

Result: The domain's publicly available DNS records, IP information, related hosting information, and network ownership details were examined.

Experiment: Armitage

week 9 lab manual_armitage.pdf

1. Aim

To use Armitage, a graphical front end for Metasploit, to organize lab targets, discover hosts, identify relevant modules, and examine sessions in an authorized lab.

2. Procedure

Step 1: Open Kali Linux

1. Open Oracle VM VirtualBox.
2. Select Kali Linux and click Start.
3. Log in if prompted.
4. Open Terminal.

Step 2: Check whether Armitage is installed

Type these commands separately:

```
armitage --help
msfconsole -v
```

If the commands display help and version information, the tools are available.

Step 3: Install Armitage if required

```
sudo apt update
sudo apt install armitage
```

Wait for installation to finish. Skip this step if Armitage is already installed.

Step 4: Start PostgreSQL

```
sudo service postgresql start
```

Initialize the Metasploit database on first setup only:

```
sudo msfdb init
```

Check the database status:

```
sudo msfdb status
```

Step 5: Launch Armitage

```
sudo armitage
```

When the connection window appears, your PDF lists these default settings:

Field	Value
Host	`127.0.0.1`
Port	`55553`
User	`msf`

1. Keep the defaults unless your lab specifies otherwise.
2. Click Connect.
3. If asked whether Armitage should start the RPC server, choose Yes.
4. Wait for the main window.

Step 6: Create a workspace

1. Click Workspaces → Manage.
2. Create a workspace named `lab-week9-armitage`.
3. Create it and switch to it.

A workspace organizes the data for your practical.

Step 7: Discover hosts

1. Click Hosts → Nmap Scan.
2. Select Quick Scan (OS detect).
3. Enter the authorized lab network range given by your instructor.
4. Start the scan.
5. Wait for discovered hosts to appear in the Targets panel.

Step 8: View host services

1. Right-click an authorized host.
2. Select Services.
3. Review the recorded ports and services.

Step 9: Find attacks

1. Select the authorized lab host.
2. Click Attacks → Find Attacks.
3. Wait for Armitage to identify potentially relevant modules.
4. Review the suggestions.

A suggested module does not guarantee that the host is vulnerable. Only continue with an exploit if it is explicitly part of your instructor's authorized lab exercise.

Step 10: Understand Meterpreter

If an authorized test establishes a session, your PDF describes interacting with it through Meterpreter → Interact.

The commands demonstrated in the manual include:

`sysinfo`

Displays system information.

`getuid`

Displays the session's user identity.

`ps`

Lists processes visible to the session.

Step 11: Troubleshooting

Problem	Action
Armitage cannot connect	Check `sudo msfdb status` and start PostgreSQL.
Connection refused	Wait 30–60 seconds and try again.
Find Attacks is slow	The first run may take longer while modules are indexed.
No session created	Record the error and ask the instructor to verify the lab configuration.

Result

Armitage was connected to Metasploit, a workspace was created, authorized hosts were discovered, and available modules were examined.

Experiment: Nikto

Week-4 Nikto.pdf

1. Aim

To scan an authorized web server for potential vulnerabilities and common security misconfigurations using Nikto.

2. Procedure

Step 1: Open Kali Linux

1. Start VirtualBox.
2. Start Kali Linux.
3. Open Terminal.

Step 2: Check whether Nikto is installed

```
nikto -Version
```

If a version is displayed, continue. Otherwise, install it.

Step 3: Install Nikto if necessary

```
sudo apt update
sudo apt install nikto -y
```

Wait for the installation to complete.

Step 4: Identify the target IP

Use the IP address of the authorized training web server provided by your instructor.

Your PDF uses `192.168.56.10` as an example. Replace it with your assigned target if different.

Step 5: Run a basic scan

```
nikto -h 192.168.56.10
```

Meaning:

- `nikto` starts the scanner.
- `-h` specifies the target host.
- `192.168.56.10` is the example target address.

Step 6: Wait for results

Allow the scan to finish. Examine the terminal output for:

- Web server information.
- Potentially outdated software.
- Interesting files or directories.
- Missing security headers.
- Other reported configuration issues.

Not every finding indicates an exploitable vulnerability.

Step 7: Save the output

```
nikto -h 192.168.56.10 -o report.txt
```

This saves the scan output to `report.txt` in the current working directory.

Step 8: Understand common options

Option	Purpose
`-h`	Specify target host
`-p`	Specify port
`-ssl`	Force SSL/HTTPS
`-o`	Save output
`-Format`	Choose output format
`-Tuning`	Restrict scan categories

Result

The authorized web server was scanned using Nikto, and the findings were examined for potential vulnerabilities and misconfigurations.

Experiment: nslookup

Week-4 Nikto.pdf

1. Aim

To query DNS records and resolve domain names into IP addresses using `nslookup`.

2. Procedure

Step 1: Open Kali Linux

1. Start Kali Linux.
2. Open Terminal.

Step 2: Check whether nslookup is available

Type:

```
nslookup
```

If an interactive prompt appears, the tool is available. Type `exit` to leave the prompt.

If it is missing, install the DNS utilities package:

```
sudo apt update
sudo apt install dnsutils
```

Step 3: Find a domain's IP address

```
nslookup example.com
```

Press Enter.

Look at the response for the DNS server used and the returned address records.

Step 4: Find mail servers

```
nslookup -type=mx example.com
```

The `-type=mx` option requests mail-exchange records.

Step 5: Find nameservers

```
nslookup -type=ns example.com
```

The `-type=ns` option requests nameserver records.

Step 6: Find TXT records

```
nslookup -type=txt example.com
```

TXT records can contain domain-verification information and email-security policies.

Step 7: Query a particular DNS server

```
nslookup example.com 8.8.8.8
```

This asks the DNS server at `8.8.8.8` to resolve the domain.

Step 8: Perform a reverse lookup

```
nslookup 8.8.8.8
```

This attempts to find a hostname associated with that IP address. A reverse record may not exist.

Step 9: Record your observations

Write down:

1. The domain queried.
2. The IP address returned.
3. The MX and NS records, if available.
4. The DNS server used.
5. Any errors or records that were not returned.

Result

The `nslookup` utility was used to query domain names, inspect DNS records, and examine IP-to-hostname resolution.

Tool	Simple purpose	
1	VirtualBox	Runs virtual computers inside your computer.
2	Kali Linux	An operating system containing tools for ethical hacking and security testing.
3	Metasploitable 2	An intentionally vulnerable machine used to practise security testing.
4	Windows XP	A target operating system used for lab practice.
5	Sysmon	Monitors and records system activities and events in Windows.
6	theHarvester	Collects publicly available information about a domain, such as email addresses and subdomains.
7	Netdiscover	Discovers devices connected to a local network.
8	WHOIS	Finds domain registration information, such as registrar and registration dates.
9	nslookup	Finds DNS information, including a domain's IP address.
10	DNS Mapping	Identifies the DNS records and subdomains associated with a domain.
11	Nikto	Scans web servers for potential security issues and misconfigurations.
12	Google Dorking	Uses advanced Google search operators to find specific publicly indexed information.
13	FOCA	Extracts metadata from documents to discover information about their authors and software.
14	Maltego	Creates visual graphs showing relationships between domains, IP addresses, people, and other entities.
15	Shodan	Searches for Internet-connected devices and their exposed services.
16	Nmap	Discovers network hosts and scans their ports, services, and sometimes operating systems.
17	Nessus	Scans systems and networks for known vulnerabilities.
18	Robtex	Provides information about domains, IP addresses, DNS records, and network relationships.
19	Metasploit Framework	Provides modules for testing vulnerabilities and validating security weaknesses.

20	Armitage	Provides a graphical interface for using Metasploit.

Quick revision for your viva

- Information gathering: theHarvester, WHOIS, Google Dorking, Shodan, Robtex.
- Network discovery: Netdiscover, Nmap.
- DNS lookup: nslookup.
- Web server scanning: Nikto.
- Document metadata: FOCA.
- Visual relationship mapping: Maltego.
- Vulnerability scanning: Nessus.
- Penetration testing: Metasploit, Armitage.
- Windows activity monitoring: Sysmon.
- Virtual lab setup: VirtualBox, Kali Linux, Metasploitable 2, Windows XP.